

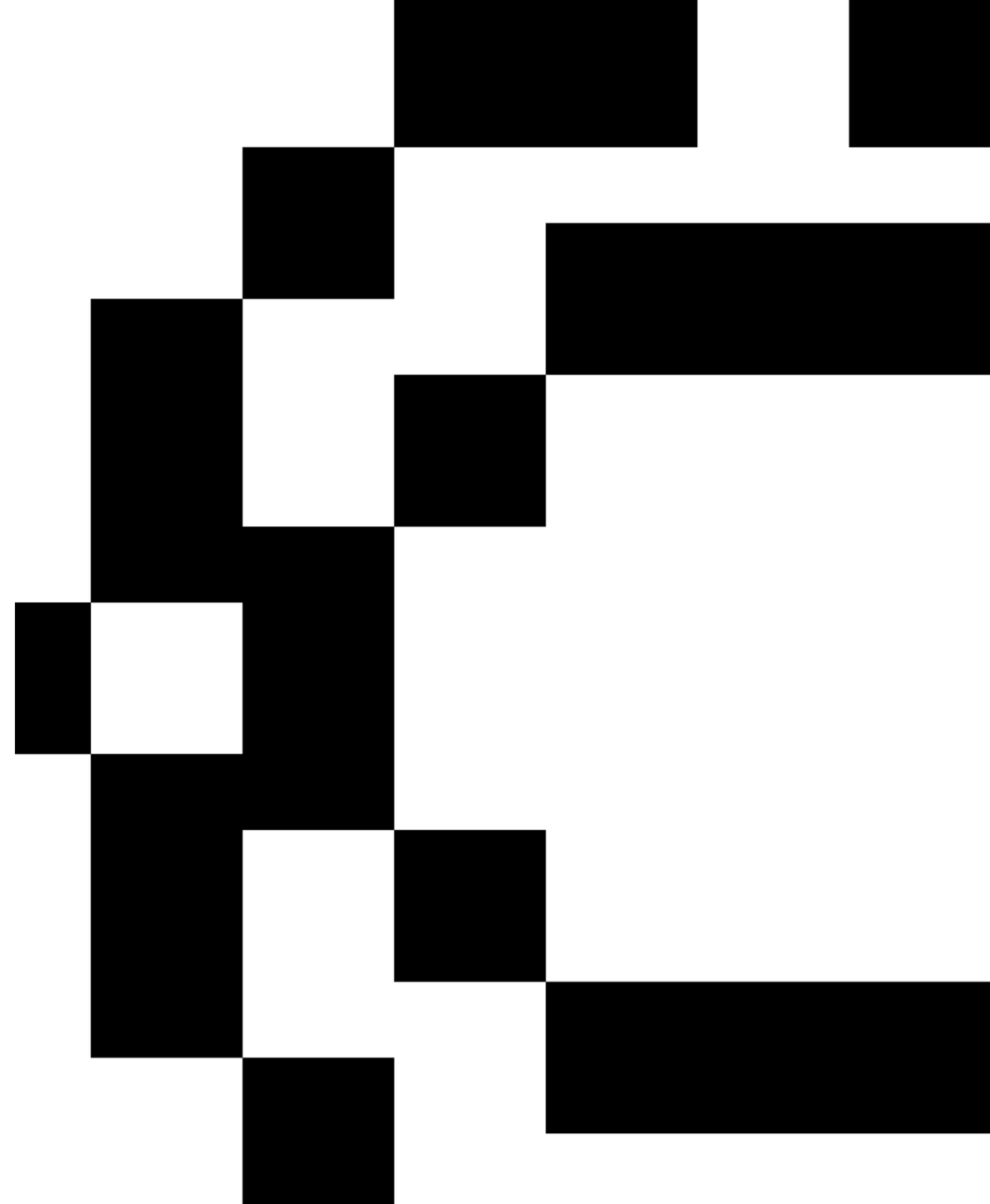
Olist

Supply-Chain

Risk Intelligence

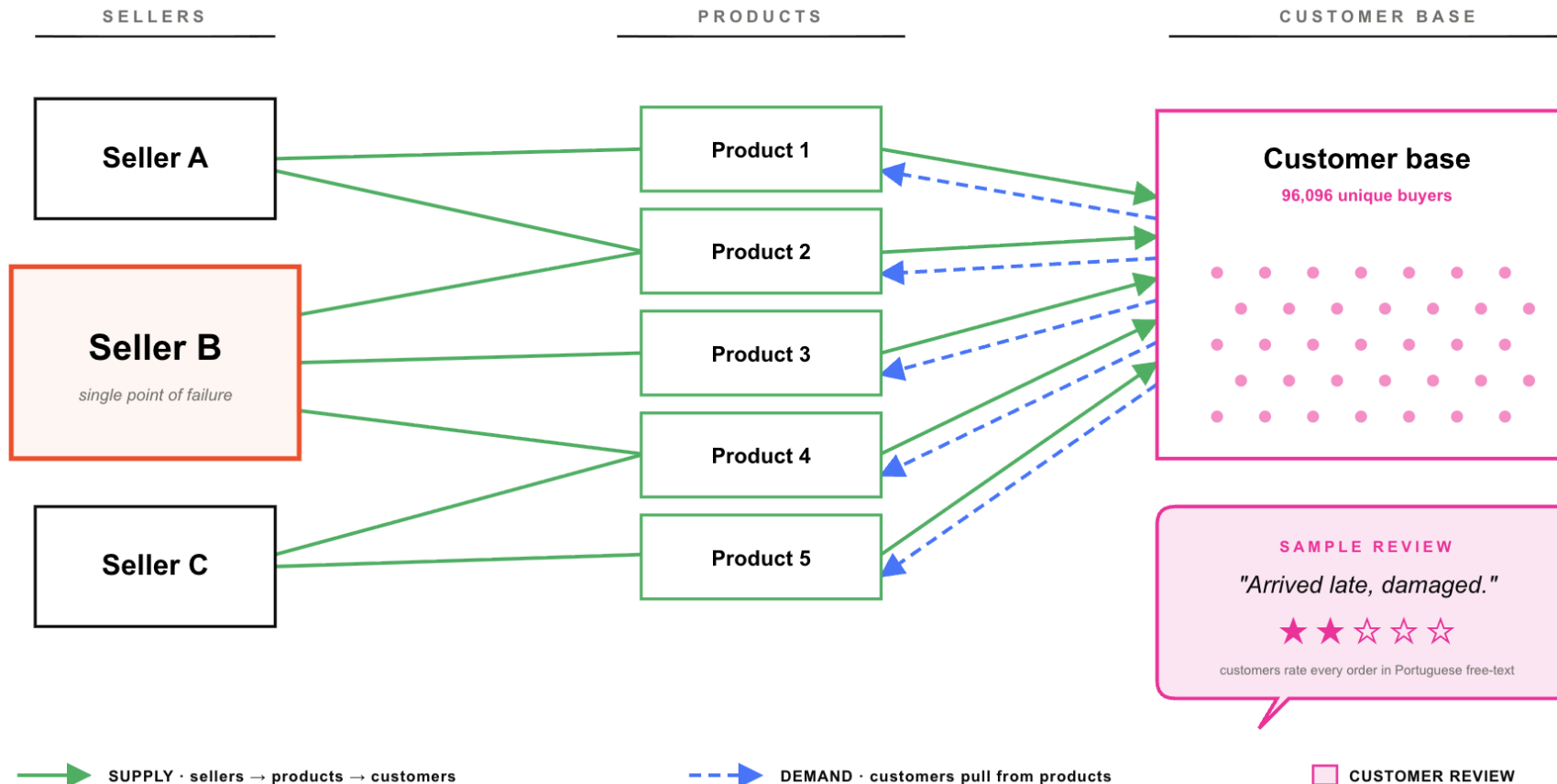
A **project update** for Olist Management

Lukas Belser, Samuel Braun, Margarida Quintino, Jan Their
Big Data Analytics – 12.06.2026



Top sellers carry the marketplace, Olist sees their silent failure last

Which sellers should Olist act on?



32.2%

of revenue concentrated in the top 50 sellers (of 3,095)

1,790

customers depend on a single top seller
· 1.86% of the base

104,162

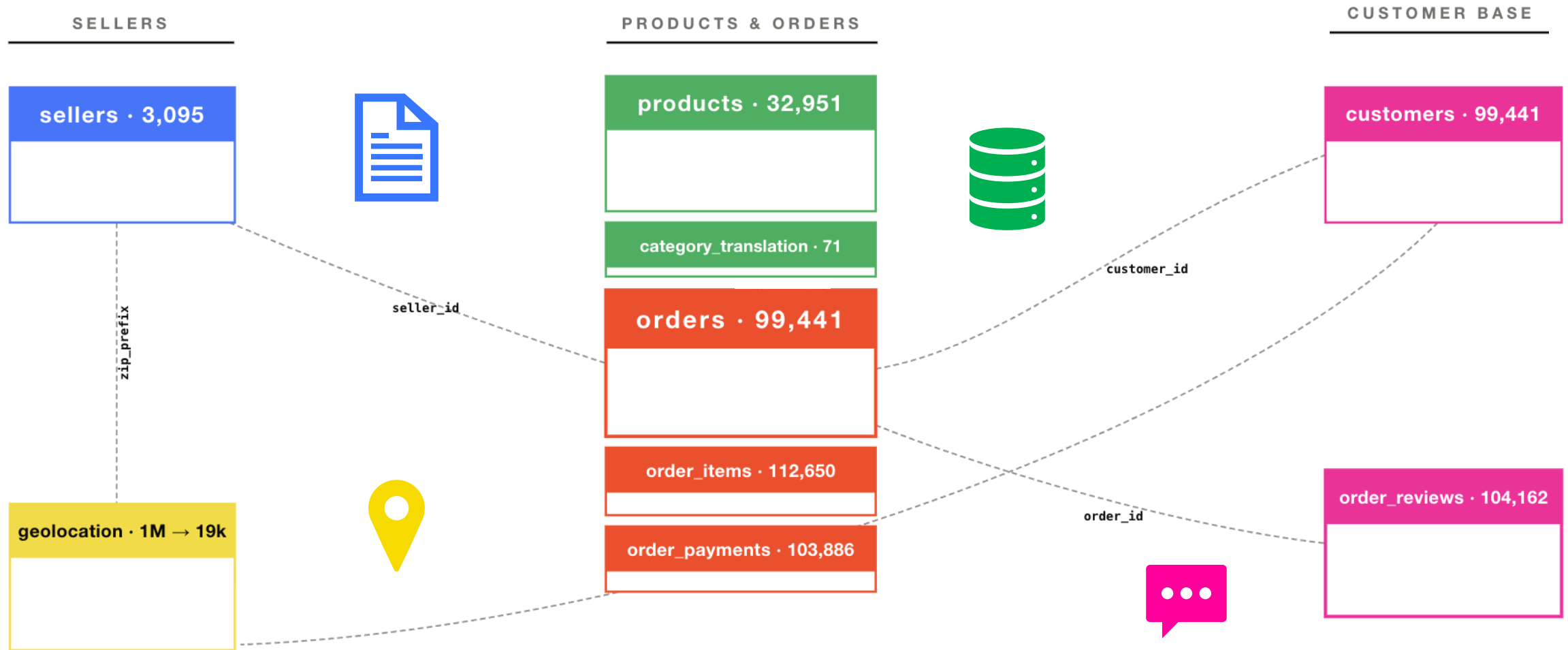
customer reviews · 43k carry Portuguese free-text complaints

99,441

orders across 2 years (Sep 2016 to Aug 2018) · 96,476 delivered

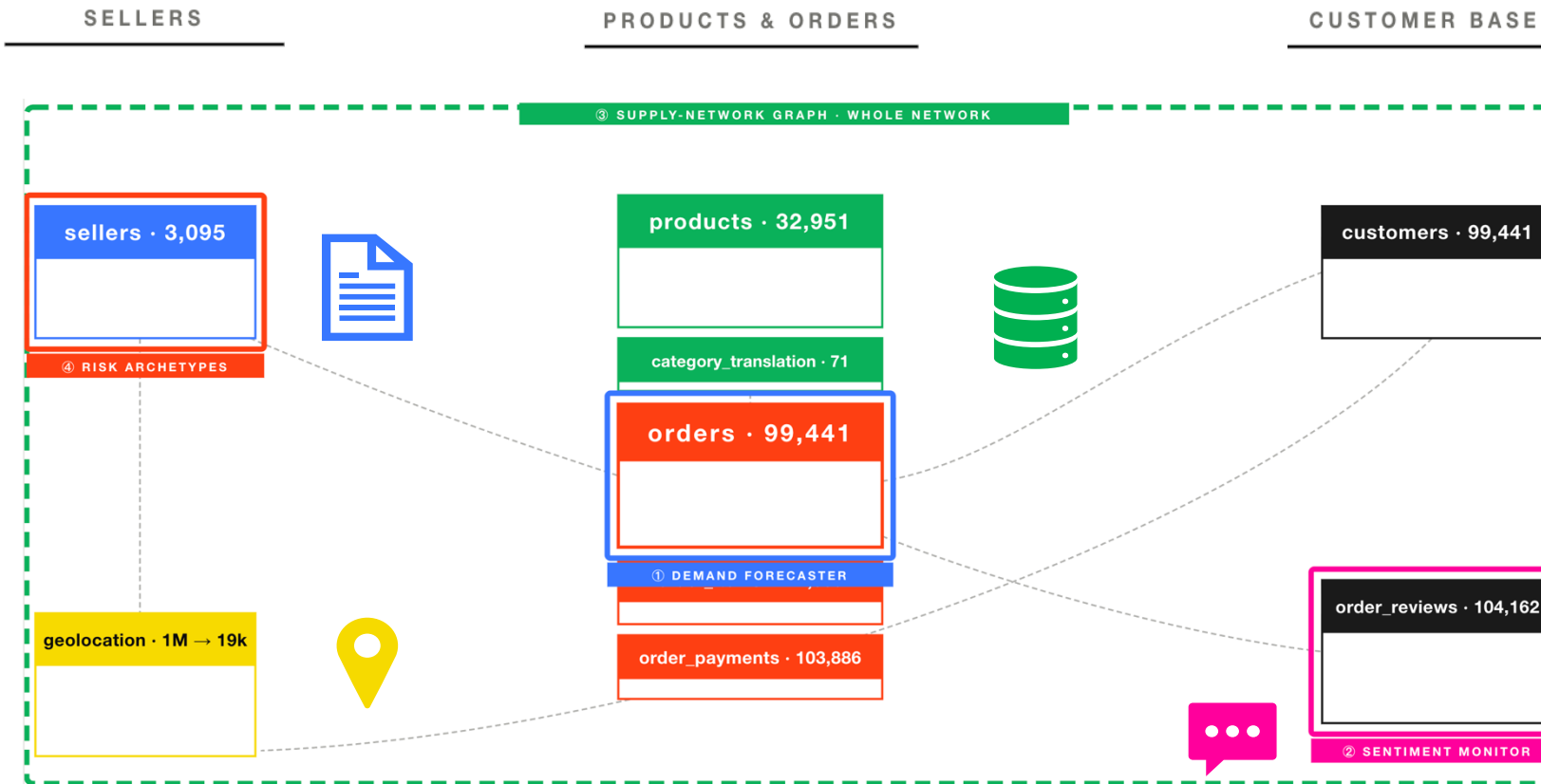
Multiple distributed data sources, value comes from joining them

9 data sources



Three risk signals, one metric: Seller Risk Index

Four prototypes light up the data network



① Demand Forecaster

Forecasts each seller's weekly order volume and tracks delivery lateness (Spark MLlib). The lateness signal feeds the risk index; the growth forecast flags **sellers outgrowing their delivery capacity**.

② Sentiment Monitor

Reads every Portuguese review and scores customer satisfaction per seller, reliably separating complaints from praise (Spark NLP pipeline plus a PyTorch deep-learning model). Satisfaction is a **direct health signal**, not a volume predictor.

③ Supply-Network Graph

Maps which sellers share customers (GraphFrames) to find structural single-points-of-failure and name a **backup seller** for each. This is risk no per-seller dashboard can see.

▼ CONVERGE

SELLER RISK INDEX

SAFE · WARNING · CRITICAL

per seller · weekly · archetype + action

④ Risk Archetypes

Spark MLlib groups sellers into four failure modes (**delivery-driven**, **satisfaction-driven**, **structurally-critical**, **low-risk**), so each can get a targeted response.

Even at today's volume, the marketplace produces the 4 V's

All four show up structurally, not because of dataset size

V₁ Volume

~1.5M rows

99,441 orders · 112,650 order-line items · 104,162 reviews · 103,886 payments · 1M raw geo points. Today's dataset spans **2 years**; the marketplace produces this every quarter at scale.

V₂ Velocity

weekly refresh

New orders, reviews, and delivery events arrive **continuously**. The Seller Risk Index rescores every seller weekly, running the same code every Monday.

V₃ Variety

4 data shapes

Structured orders + **Portuguese free-text reviews** (43,337 rows) + geographic coordinates (19,015 centroids) + graph relationships (seller↔customer). One Spark pipeline absorbs all four.

V₄ Veracity

dirty in → trusted out

Raw marketplace data is **incomplete and noisy**, and Spark cleans it before any seller is scored: **~2,965 orders with missing delivery dates** are dropped (their timing can't be trusted); **~1M scattered geo-points** collapse to 19,015 reliable zip-centroids; ambiguous neutral reviews and a tail of extreme delivery outliers are filtered out. The watchlist only ever sees data the client can **trust**.

Four challenges, each met with the matching Spark tool

The dataset is small today, but the problem is big-data-shaped, so the tooling has to be too

BIG-DATA CHALLENGE	THE RIGHT SPARK TOOL	WHAT IT MEANS FOR OLIST
V₁ Volume 9 tables, ~1.5M rows, joined every run and growing	Distributed joins , with the small lookup tables broadcast to every node	Adds machines, not rewrites. The same pipeline runs when the marketplace is 10× bigger.
V₃ Variety orders, Portuguese reviews, maps, and the seller↔customer network	One Spark stack for SQL, MLlib, a deep-learning model, and GraphFrames	One pipeline reads all four; the graph answers a structural question no spreadsheet can.
V₂ Velocity orders and reviews never stop arriving	Weekly batch scoring , plus a Structured Streaming layer for live signals.	Batch produces the score (needs full history); streaming flags spikes between runs.
V₄ Veracity raw data is incomplete and noisy	Spark cleaning that drops, dedupes, aggregates, and caps outliers.	Every seller score is built only on data the client can trust.

~1.5M → ~15M rows

SAME PIPELINE · 10× THE DATA

Every step runs **across a cluster**, so nothing assumes the data fits on one machine. Scaling 10× adds hardware, not a rewrite.

Three things a per-seller dashboard cannot tell you

Each finding comes from combining signals, not from any single seller KPI

01 · SIGNAL INDEPENDENCE

Risk is **three separate problems**, not one.

Demand pressure, customer sentiment and network criticality barely move together. The strongest correlation between any two is just 0.21. A seller can ship on time, keep customers happy and still be a structural single point of failure. **No single KPI catches all three.**

02 · SUPPLY-NETWORK GRAPH

The biggest sellers aren't the **irreplaceable** ones.

Ranking by size just re-finds the biggest accounts. Our substitutability signal surfaces **small sellers no competitor can cover**, invisible to a volume dashboard, yet the ones whose failure the marketplace cannot absorb. 13 at-risk sellers have no backup anywhere.

03 · CUSTOMER SENTIMENT

Reviews **confirm** trouble, they don't **predict** it.

Our model separates happy from unhappy reviews almost perfectly, yet sentiment moves almost in lockstep with sales rather than ahead of them. **Delivery delay moves first.** So Olist should trigger on delays and use sentiment to confirm, never the other way round.

WHY WE BUILT ONE NUMBER

The three signals are independent, so no single metric ranks risk alone. We fold delivery, satisfaction and structural importance into **one weighted score per seller.**

NEXT SLIDE

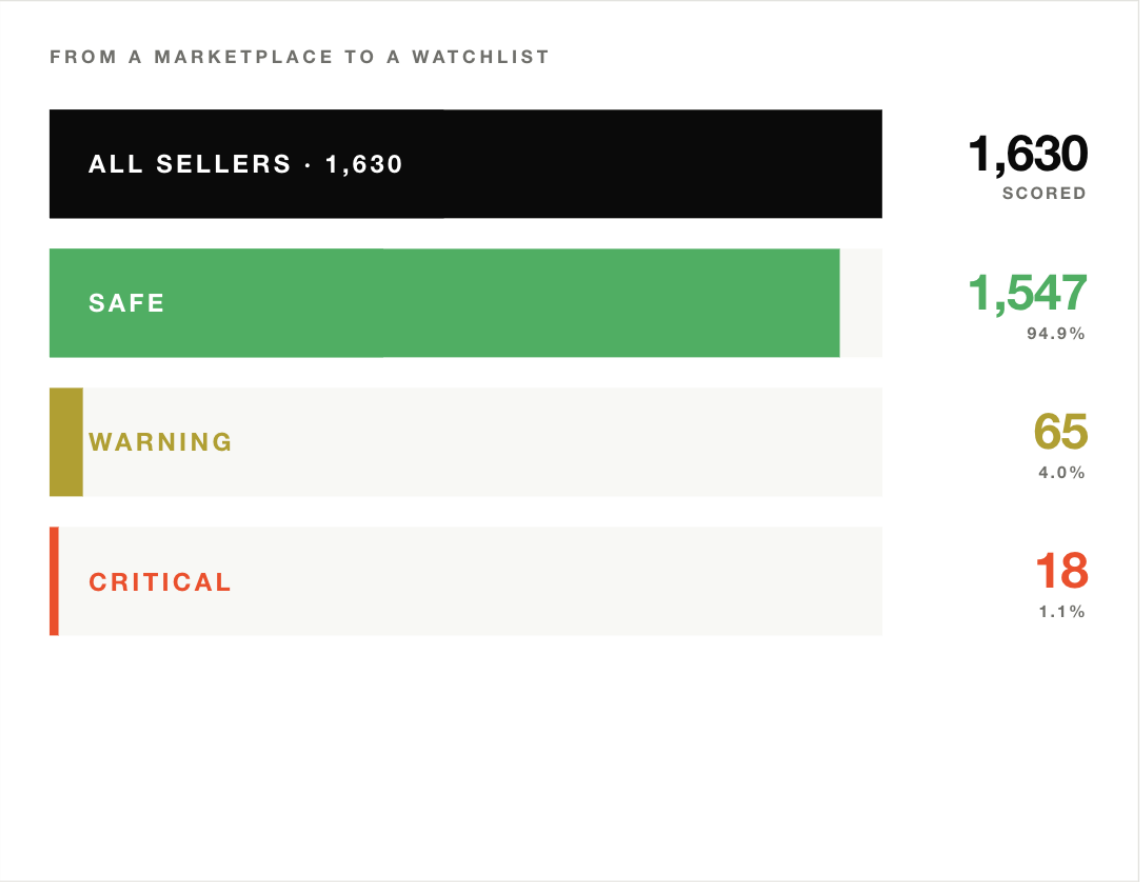


The Seller Risk Index

1,630 sellers → 83 weekly calls

Risk Index distils 1,630 sellers into 83 calls per week

Three weighted scores → one banded composite → weekly watchlist with archetype & recommended actions



WEEKLY SELLER WATCHLIST					WEEK OF 2026-05-25
SELLER	ST.	BAND	ARCHETYPE	RECOMMENDED ACTION	
54965bbe...	PR	CRIT	Delay-driven	Logistics audit	
b1b39487...	SP	CRIT	Sentiment-driven	Account-manager call	
e64d65bc...	SP	CRIT	Sentiment-driven	Escalate — no backup	
b5abf4f3...	DF	CRIT	Sentiment-driven	Account-manager call	
7daca083...	SP	WARN	Delay-driven	Escalate — no backup	
1b45169b...	PR	SAFE	Best in class	—	

From post-complaint discovery to an 83-seller weekly watchlist

Four valuable implications.

01 · POSTURE

REACTIVE → PROACTIVE

Weekly risk score per seller

No more waiting for refund tickets. Every seller is rescored every Monday and assigned a band and archetype, **before the customer complains**.

1,630

SELLERS SCORED · EVERY WEEK

02 · WORKLOAD

1,630 SELLERS → 83 CALLS

Actionable shortlist

Account managers get a short watchlist (**5% of the portfolio**) with a recommended action per archetype.

83

18 CRITICAL + 65 WARNING THIS WEEK

03 · RESILIENCE

SINGLE POINT OF FAILURE → NAMED BACKUP

Backup seller per hub

The supply graph names a backup for **~1,520 sellers** and flags the **13 with no backup anywhere** to escalate first. A hub outage **no longer becomes a category outage**.

~1,520

SELLERS WITH A NAMED BACKUP · 13 ESCALATED

04 · SCALE

TODAY'S CSVS → TOMORROW'S CLUSTER

Same pipeline scales 100×

PySpark code runs on a laptop today and on Databricks / EMR tomorrow. **The prototype IS the production path.**

0

LINES OF CODE TO REWRITE FOR PRODUCTION

Productionise prototypes on Databricks

What we built, and whats next

WHAT WE BUILT

- One reproducible **PySpark notebook** with visible outputs in every cell, deterministic across re-runs
- **Four prototypes**: Demand · Sentiment · Supply-Network Graph · Risk Archetypes
- One unified deliverable: the **Seller Risk Index**, a weekly watchlist with archetype and action
- A **Structured Streaming** prototype re-expressing weekly scoring as a live stream (bonus)

WHAT'S NEXT

- **Calibrate and back-test the Index** against realised churn/failure, then tune weights and band cut-offs
- Productionise on **Databricks / EMR** as a scheduled weekly Spark job
- Wire the watchlist into the **seller-success CRM**, with Slack alerts on new WARNING entries
- **Expand the graph**: add category, postal region and shipper as new vertex types

From a list of sellers to a **managed network**, and from post-complaint discovery to **weekly intervention**.

Thank you

QUESTIONS WELCOME

REFERENCES & SOURCES

Dataset: Olist (2018), *Brazilian E-Commerce Public Dataset*, Kaggle (CC BY-NC-SA 4.0): ~100k orders, 2016 to 2018, 9 relational tables.

Frameworks: Apache Spark 3.5 / PySpark · GraphFrames 0.8.3 · PyTorch 2.8 (TorchDistributor) · Plotly.

Methods: PageRank (Page et al. 1999) · Label propagation (Raghavan 2007) · Gradient boosting (Friedman 2001) · LSTM (Hochreiter & Schmidhuber 1997).